

What a Complete Action for a Λ -Tied MOND Law Would Need: An Obstruction Map and a Frozen Gate List

The authors — draft for review, 2026-09-28, version 1 (brought up to date through CFG67 and the ChainCert library; CFG68 to CFG74 are not covered except where cited: the Λ CDM comparator of CFG69, whose ultra-faint gate CFG73 and CFG74 find weakly diagnostic, is not used here) — AI-assisted research programme; not peer reviewed; nothing here is claimed closed.

Abstract

The programme ties the MOND acceleration scale to the vacuum, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived [Z26a,Z26c]. Its best current version, candidate B, is an *effective law*: no committed action produces it. This note records what one action would have to supply, and what is already known about each requirement. (i) A gate list was frozen before any assembly attempt: 63 gates in five groups. B passes 42 of 48 harness gates by recount, inferred rather than established, and is short or marginal on a further list of rows that we give with their committed sizes. (ii) Two objects are missing between B and one action: a legal, bound-only, owned switch (Gap 1), and a conserved cold fluid that *is* the law’s dark density (Gap 2), with the a_0 – Λ tie (Gap 3) partly filled. Gap 1 is a scoped no-go in every construction tried: a gate on a shift-symmetric field leaves the dynamical mass equal to the baryonic mass beyond its edge (Gauss), ownership is acausal inside a reduced single-field action or a prescribed label (a causal doubled-field version returns the same reaction), and the enclosed-mass exchange reacts on the baryons. Local gates fail a second-variation stability test on 44 of 48 cases, nonlocal enclosed-mass and top-level-ball gates pass 48 of 48, and a dynamical gate scalar is stable only with two new untied constants and DE13’s own cost. Gap 2’s target, $C(r) = \rho_c r^3 g_{\text{tot}} = (a_0/4\pi) M_b(< r)$, is excluded as an equation of state, a local closure, a second force, a constraint action, a coupling to the phantom potential, a density slaving and a tidal-tensor coupling, each under stated hypotheses. Gap 1 and Gap 2 reduce to one non-derivable object (as an identification, not a theorem), and a documentary count of 54 committed constructions (48 construction rows plus six screened-out rows) finds none that derives it (D=0, P=11, I=5, X=27, U=11; U=5 without the screened-out rows). The tie can be written into a conserved fluid’s stress cap in one action without a new local degree of freedom, but a barotropic saturating cap engages over at most about $11\times$ in mass against the 10^4 the baryonic Tully–Fisher relation needs (a scoped screening result; the referee’s caveat that this is not a ceiling is given). (iii) The derived cold-mass rule closes the ultra-faint offset (from $+0.325$ dex to -0.06 dex, a weak test by the referee’s account) but over-predicts the classical satellites (M31 LVD -2.67σ); calibrated on the galaxies’ own inner kinematics it leaves the SLUGGS early types 2.6σ short; no one of four pre-declared readings passes all seven gates, and no single fraction of its debris reconciles ten populations at a 1σ -per-population acceptance (at 2σ CFG59’s ten populations do share $\varphi \approx 0.5$ – 0.7 , CFG75; with SLUGGS’s dynamical masses only SLUGGS blocks the 2σ intersection). These dwarf verdicts are kernel-robust, robust in size to the concentration relation (though marginal in significance, CFG65) and to an Einasto shape, and a Burkert core removes both the closure and the over-prediction. (iv) Lean certifies 161 theorems: that stated conclusions follow from stated premises, no empirical fact. (v) What decides the rest: Gaia DR4 wide binaries (2 December 2026), which can separate B only from the bare MOND law; a_0 at $z \approx 2.5$ (not testable with the data on disk; the PHIBSS sample is empty under the pre-registered selection); and multi-epoch ultra-faint data. The document is a map, not a result.

1 The frozen gate list, and where candidate B stands on it

Provenance. The gate list (`campaign_fresh_gravity/closure_map/GATES.md`) was compiled on 2026-09-28 by a delegated reader of the committed repository and spot-checked by hand; where it disagrees with a committed script output, the script wins. It was frozen before any assembly attempt, so an assembly cannot move the goalposts. Footings: canonical $a_0 = 9.3603 \times 10^{-11}$ and alternative $a_0 = 1.1312 \times 10^{-10} \text{ ms}^{-2}$; pairs are written canonical/alternative; $\kappa = \frac{1}{2}$ is fitted throughout.

Group counts. Recounted here from the tables of `GATES.md` (an agent-derived count, re-run by the audit script):

group	gates	content
(1) Galactic dynamics	24	SPARC, dwarfs, ellipticals, lenses, binaries
(2) Groups and clusters	5	X-COP, Bullet, X-ray groups, Local Group
(3) Cosmology and lensing	13	CMB, growth, KiDS, budget, $a_0(z)$, forest
(4) Local and relativistic	8	Cassini, GW170817, PPN, wide binaries
(5) Theory consistency	13	stability, causality, Noether, FRW (the C-H/K branch)
total	63	

Model-independent versus soft. Each row carries the data-side class assigned by the programme’s evidence audit (CFG1): MI (model-independent: the systematics budget is below the stake), SOFT (budget at or above the stake) and CONT (published analyses disagree); “n.c.” means not classed. That audit had 33 results, of which 14 were MI, 11 SOFT and 8 CONT. Counting the class column of the 63 rows (recounted here) gives MI on 14, SOFT on 10, CONT on 9, n.c. on 13, and no class on 17 (the 13 theory rows and 4 others). “n.c.” is a statement about the audit’s coverage, not a finding that a row is model-independent; the lanes run after the audit (for example the ultra-faint referee) are unclassified.

The harness and its thresholds. The scored harness has 52 rows, of which 48 are scored and 4 are “BUDGET strict (reported)” rows that are not counted. Thresholds are the harness’s own: SPARC rms ≤ 0.110 dex; KiDS $\Delta\chi^2 \leq +9$; lenient budget $\Omega_{\text{ph}} \leq \Omega_c = 0.265$; X-COP identity within 20%; Bullet collisionless mass $> 2\times$ the aperture baryons; CMB-lensing $|\text{pull}| \leq 2\sigma$; forest deviation ≤ 0.10 ; every population gate $\leq 2.0\sigma$. B’s own ledger in that harness is: fitted 2 (κ , $\Omega_c h^2$), declared 5 (kernel shape, edge $x_e = 0.4$, the max rule T5, the switch criterion, the ownership rule), tied 1, derived 1 (Solar-System screening). The 6 failing scored rows are 3 distinct gates times 2 footings.

The count. Candidate B passes 42 of 48 harness gates *by recount* (lane CFG19 substitutes four M31 rows). The harness’s own control K2 failed, and X-COP and the Bullet cannot be re-scored under the derived cold-mass rule without circularity (CFG39), so 42 of 48 is inferred, not established. The strict cold-mass budget fails on all four cells (1.4–1.7 $\times$), outside the 48.

Where B fails or is marginal. Sizes are the committed ones; “can/alt” are the two footings.

id	observable	B’s standing	class
1.09	MW ultra-faints	FAIL: 7.97/7.52 σ in the harness (statistical only); 3.8/3.5 σ refereed (+0.325/+0.304 dex, Kaplan–Meier with a 0.077-dex floor)	n.c.
1.15	Chae external-field signal	FAIL: 4.08/4.29 σ (Chae’s fits); refit under the programme’s law 1.7 (P2, can), 2.7 (P2, alt), 2.2 (ν_{mono} , can), 3.0 (ν_{mono} , alt) σ	SOFT
3.06	strict cold budget	FAIL on all four cells: 0.229/0.230 (can), 0.264/0.266 (alt) against 0.159; reported, not counted	n.c.
2.03	20 X-ray groups	R_{500} marginal, 1.41/1.33 (1.80/1.49 σ); R_{2500} FAIL, 1.88 (2.57/2.63 σ); shortfall tracks the baryon fraction, $\rho = -0.958$	n.c.
2.04	Local Group R_0	FAIL: 3.0–4.8 σ high at the KiDS floor; not framework-specific (standard halos in the same machinery: $T = 24.5\text{--}72.5$ against 26–28)	MI marginal
3.07	budget against KiDS	FAIL as declared: 29.4/32.9 (can), 45.4/49.3 (alt); “not decidable” (smallest standard-halo spread $S = 50.2$)	SOFT
1.17	binary galaxies	undecided: circular orbits $A = 1.511/1.443$; B’s timing orbits 1.116/1.047 pass, the profile ($-4.2/ -4.1\sigma$) does not	n.c.
1.18	X-ray ellipticals (7)	marginal/fail: 1.70/1.58 σ (factor 1.91/1.79)	n.c.
1.19	SLACS lenses	marginal: 1.5–1.7 σ with a 0.10-dex floor, 6.9–7.8 σ statistical	CONT
1.22	UGC 2487 under the cold-mass rule	FAIL on one galaxy: +0.139 dex	n.c.
1.08, 1.10, 1.11, 1.16	M31 dSphs; LV dwarfs; cluster-infall zero point; Coma UDGs	marginal passes: 1.63/1.24 and 1.44/1.21 σ ; 1.71; 1.30; 1.33/1.11 (2.30/2.08 with stars only)	SOFT / MI / SOFT / MI
5.01–5.02	one action; stability	the one covariant action on record (V0) is not complete: its region gate is obstructed when varied; the gate’s second variation lands on the baryons, $c_{\text{gate}} = 1526\text{--}3741 \text{ km s}^{-1}$ at $z = 0.25$ against gas at 37/117 km s^{-1}	—

Two further items sit outside the frozen list. The most massive star-forming disks (23 super spirals, CFG40) run +0.11 dex fast (1.4 σ), the nine fastest +0.17 dex (2.06 σ), and the declared test fails at the margin; with Simard’s measured bulge fractions (CFG56) the nine fastest are +0.164 dex (2.34 σ), so the baryon structure model is not a way out. Not scorable until data exist are Gaia DR4 wide binaries and a_0 at $z \approx 2.5$ (Section 7).

Bulge fractions (CFG56). CFG40 put all the baryons in a disc, and its referee found that a point mass passes. CFG56 repeats the test with Simard et al.’s measured bulge-to-total ratios (0.04–0.46, median 0.26) and CFG40’s

thresholds. Canonical offsets: mean $+0.105 \pm 0.063$ (1.67σ), slope against $\log M_b$ 1.80σ , the nine fastest $+0.164$ (2.34σ). The clause that the bare law fits within 2σ (H1) passes and H2 fails on the nine-fastest clause. The model floor shrinks from 0.041 to 0.001 dex, so the significance rises rather than falls, and $f_{\text{ex}} = 0$ in all 23, so the cold-mass rule cannot cure it; a point mass is not a fair model of a disc with $B/T \approx 0.26$. Under the P2 kernel the mean offset is $+0.132$ (2.06σ) and H1 itself flips to fail (CFG64; Section 5.1). The fail stays marginal. An independent re-derivation (CFG89) reproduces these numbers and shows that the failing nine-fastest clause is selected on the observed speed and adds almost nothing beyond the 23-mean (selection alone gives an excess of $+0.071 \pm 0.016$ against the observed $+0.059$; the unselected test is the slope clause at 1.80σ), and that the verdict is decided by the stellar-mass floor convention (a $+0.1$ dex stellar-mass shift alone passes it).

Status refresh since the freeze. A separate status file (`campaign_fresh_gravity/closure_map/GATES_STATUS_2026-09-29.md`) restates every row against the lanes committed after the freeze, without moving the list, its thresholds or its ids. Its bottom line: “Candidate B’s gate record has not improved since the freeze. It has sharpened.” The one clean pass of B’s derived rule (SLUGGS) did not survive dynamical stellar masses (Section 5); the rule closes the ultra-faint failure but breaks the classical satellites and the Local Volume field dwarfs; and at the action level the gaps are scoped no-gos (Section 3).

Reading the table. Of the rows above, the SOFT and CONT classes (1.15, 3.07, 1.19, 1.08, 1.11) mean that the data-side systematics are comparable to the stake, so the size of the failure is not decisive by itself. The ultra-faints (1.09) are the failure the programme’s own lane README calls B’s largest standing one; tides, noise and unresolved binaries do not explain it (CFG28, CFG29). None of these failures is a theorem about a complete action; they are gates any action would still have to pass on top of B.

2 The effective law B, and what an action must supply

Candidate B is CFG4’s target law (T1–T6) plus a hierarchical ownership rule (FG001). It has no action.

- **T1:** $a_0 = \kappa c \sqrt{G\rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted; the form is unique, κ is not derived.
- **T2:** $g = \nu_{\text{mono}}(g_{\text{bar}}/a_0) g_{\text{bar}}$ inside “on” regions.
- **T3:** a region is on iff (i) it is a top-level bound system that has turned around, (ii) a density edge $x_e \in [0.31, 0.48] r_{\text{ta}}$ (declared 0.4), and (iii) $M_b \geq M_*$.
- **T4:** a cold fluid with $\Omega_c h^2 = 0.12$, fitted as in Λ CDM.
- **T5:** the dark mass is $\max(M_{\text{ph}}, (\Omega_c/\Omega_b)M_b)$; the max rule is declared.
- **T6:** a late allowance.
- **Ownership:** top-level systems only; accreted systems keep the isolated law of their infall baryons; systems formed embedded are Newtonian.

An action that produces B has to supply, at least:

- (a) the tie of a_0 to Λ , with κ entering as a datum;
- (b) the kernel and the galaxy law inside on regions;
- (c) a switch that is a legal action term: bound-only, top-level-owned, baryon-reading, frame-covariant, with a source-confined edge (Gap 1);
- (d) a conserved fluid whose bound-region density *is* the law’s dark density, and is plain CDM elsewhere, without double counting (Gap 2);
- (e) one action carrying both the tie and the fluid, with post-Newtonian and cosmological sectors closed (Gap 3);
- (f) the stability, causality and conservation requirements of group (5) of the gate list.

What exists. Its pieces sit in three separate places. The derivation-chain action holds the tie and a static kernel but a postulated fluid coupling, a squeezed separator and no ownership; V0 holds a region-local kernel and a baryon-reading gate but its varied gate is obstructed; the astra CA4/CA5 action holds a fluid, a heat gate and Ward/Dirac bookkeeping, but its a_0 -vacuum relation is “an optional input”, it has no bound-only switch, and every astra result is marked unreviewed. Switch constructions built on the MOND flux, on a Brown–Schutz dust turnaround gate (a ghost pole on 24/24 layers) and on generated phantoms were excluded or withdrawn. The committed verdicts are “INCOMPLETE” for each surviving row. *No committed action produces candidate B.*

3 The remaining gaps, and the reduction of Gap 2 to Gap 1

Gap map. Standing of the three gaps after the lanes CFG43 to CFG65; each entry is scoped by the hypotheses given below.

gap	what is missing	standing
Gap 1	a legal, bound-only, top-level-owned, baryon-reading, covariant, source-confined switch	scoped no-go in every construction tried: Gauss, history, the exchange’s reaction and top-level detection (CFG48); local gates unstable and nonlocal gates stable but bilocal; a dynamical scalar needs two untied constants and DE13’s cost (CFG49); neither owns the hierarchy
Gap 2	a conserved cold fluid whose density is the law’s dark density, eq. (1)	exact target; scoped no-go for every closure tried (CFG44, CFG50); reduces to Gap 1’s enclosed-mass object; no committed construction derives it (CFG60)
Gap 3	one action carrying the tie and the fluid, with PPN and cosmology closed	partly filled: the tie acts on a fluid’s stress cap with no new local degree of freedom (CFG43); the natural cap is barotropic and screens; PPN, shell crossing and the galaxy law itself are open

Gap 1: a legal switch. Missing is a gate variable equal to “inside a turned-around, top-level bound system with $M_b \geq M_*$ ” that enters as a Lagrangian term. The committed gates read the baryon flow θ_b , a density, the khronon curvature $\langle K \rangle$ or a field energy; none reads boundness or hierarchy. Ownership adds history (accreted systems keep their infall halo), which no instantaneous state variable carries. A gate that is varied as an action term is unstable: its second variation lands on the field it reads (DE12, DE13), with $c_{\text{gate}} = 1500\text{--}3700$ against gas at $37\text{--}117\text{ km s}^{-1}$, and no single smoothing length works (it needs at least 500 kpc while the flagship case holds only to 100 kpc; XR15). That obstruction assumes the gate variable is slaved to the baryon field; two later lanes relax that hypothesis, below.

Gap 1 attacked directly (CFG48). CFG48 attacks Gap 1 in six scripts, written by a lane session against action-level gates (GA–GG) that, by the lane’s own record, were written before any script (the referee’s file birth times are consistent with this, but everything was committed together, so git cannot order them), and re-run by the orchestrating session. A referee reproduced all 12 outputs from a clean export and re-derived 7/7 headline numbers with independent code. Four obstructions, each with its hypotheses in the lane README. (1) *Gauss*. A gate that multiplies a shift-symmetric MOND field whose only source is the baryons gives $M_{\text{dyn}} = M_b$ exactly beyond the edge, while B’s target is $M_b \sqrt{1 + x_e^2}$ ($15\text{--}33 M_b$); the phantom inside the edge would have to be a negative shell of 3.2×10^{11} , 2.2×10^{12} and $1.4 \times 10^{13} M_\odot$ for $M_b = 10^{10}$, 10^{11} and $10^{12} M_\odot$. The form that does keep the mass (a gate on the phantom *density*, not its flux) fails the Helmholtz test: it is not the Euler–Lagrange system of any Lagrangian in those fields. (2) *History*. Ownership is not a function of the baryon state, and a history variable inside a reduced single-field action (a one-degree-of-freedom toy; doubled-field actions are not covered by this obstruction) makes the residual at time t depend on the future trajectory; the only causal carrier there is a prescribed label, that is, ownership as initial data. CFG70 then writes the enclosed-mass exchange as a causal, reciprocal retarded Schwinger–Keldysh memory-kernel action: causality holds, but it returns CFG48’s reaction and energy demand unchanged, a scoped no. (3) *Exchange*. CFG44’s enclosed-mass energy exchange, written as a bilocal action term, reacts on the baryons outward with $0.06 g_{\text{law}}$ at $x = r/r_M = 0.3$, $0.4\text{--}0.5$ at $x = 1$ as CFG48 reports ($0.40\text{--}0.53$ in the rows of its G4 output) and $11\text{--}22$ at $x = 30$ (pass line 0.10) and must supply $23\text{--}50$ times the baryons’ orbital kinetic energy ($M_b = 10^{10}\text{--}10^{12} M_\odot$, CFG48’s convention; $23\text{--}73$ down to $10^9 M_\odot$ in CFG70, and $57\text{--}318$ with B’s committed r_{ta}). (4) *Top-level detection is not smooth*: the maximal-ball functional finds the edge with no new constant but it jumps by a factor 2 at every merger, so a variation needs a smoothing width.

An expectation that failed (CFG48, G6). The lane’s author expected every baryon-reading gate to inherit DE12/DE13’s stiffness obstruction. It does not. On DE12’s 24 layers at two widths (48 cases) the local gate fails on 44 of 48; a nonlocal enclosed-mass (Volterra) gate that reads baryon mass has no negative mode on 48 of 48 (the first negative mode would need the coupling multiplied by at least 13.7), and the rank-one top-level-ball gate is second-variation stable on 48 of 48 with either reading. The Volterra gate’s dynamical-mass reading fails on 29 of 48, every $z \leq 1$ layer. So Gap 1’s obstruction is not stability; it is Gauss, history and the exchange’s reaction. Two caveats travel with this. First, the ball gate’s first variation is an edge potential step of $0.15\text{--}3.1 v_f^2$, so DE12’s static gas is not an equilibrium of the varied theory and the second-variation pass is not a stability certificate for the full theory. Second, the referee found that this lane’s r_{ta} is not CFG4’s (CFG4’s uses the law’s phantom-inclusive enclosed mass, $1.7\text{--}3.6\times$ larger): the ball gate’s edge then sits at $0.11\text{--}0.24$ of CFG4’s r_{ta} , below

B’s window $[0.31, 0.48]$, while the no-go numbers become stronger under the committed convention. G6 exits 1 because its pre-declared instability hypotheses, 0 of 48 and 0 of 48, were false; they are kept as they fell.

A dynamical gate scalar (CFG49). The one construction that DE12, DE13 and XR36 never ran is a gate with its own stiffness: a scalar χ with kinetic coefficient μ/c^2 and mass term m_2 , sourced by a baryon-only invariant and entering through the gate $W(\chi)$. It has no ghost and is strongly hyperbolic, and both plateaus are stable for any positive μ and m_2 . Stability of the whole layer needs two new untied constants, $\mu \geq 1.5 \times 10^{28}$ to $3.9 \times 10^{28} \text{ J m}^{-1}$ and $m_2 \geq 10^{-12} \text{ Pa}$, and it then pays DE13’s own N1 cost: $\Phi_\chi(r_F) = -13.5 v_f^2$ (force $-44 g_{\text{MOND}}$, against DE13’s -31.8), $7.7 \times 10^7 v_f^2$ at the Sun, and a baryon UV speed of $2.3c$. The cost is acceptable only at $m_2 \lesssim 10^{-16}$, where 16 of 24 edges are off by more than 10% and the force is still $0.09 g_{\text{MOND}}$. The switched-stiffness repair $\mu(\chi)$ fails on its own switch-off term: no μ_0 stabilises all 24 layers at $m_2 = 10^{-12}$. Main runs: A 12/12, C 5/6, D pass and B 7/8, the one failure being a hypothesis (H2) that the run refuted and that is kept; every MUTATE control fails as required. The lane was written by a delegated agent and re-run in place. A referee reproduced nine of ten outputs from a clean export (the tenth standalone, after one non-reproducible truncation inside the run script), re-derived the reduced stability form and the $k \rightarrow 0$ theorem by hand, and re-derived the minimal stiffness against DE13’s committed value with an independent solver (ratio 0.2500 on all 32 layers; galaxy-layer maximum $3.86 \times 10^{28} \text{ J m}^{-1}$). The referee did not re-derive the finite- m_2 stiffnesses, the potential costs quoted above, CV6_C’s switched-stiffness counts or CV6_D. Process caveat: the lane has no separate frozen-criteria file, its hypotheses sit in the scripts after disclosed exploratory scans, and the files were bulk-copied, so “declared before the scripts” cannot be verified for this lane; it is below the standard of CFG48, and its declarations should be read as declared in the scripts, not as pre-registered.

Relation to CFG48. The scalar is a local field sourced by the baryon density, the class that CFG48’s G6 found unstable; CFG49 shows how far that class can be repaired: it becomes stable, but only with two untied constants and DE13’s cost. The local scalar buys a legal local action term at the price of constants; CFG48’s nonlocal gates cost no constants but are bilocal, so not a legal local term, and they carry a first-variation edge step, the Gauss lemma and the history obstruction. Neither owns the hierarchy. This is a scoped no-go for this construction class (a local scalar with constant kinetic stiffness and a baryon-trace source), not a closure of the theory, and it says nothing about the nonlocal gates.

Gap 2: the target, exactly (CFG44). Candidate B needs a cold fluid whose density is the law’s phantom,

$$\rho_c g_{\text{tot}} = \frac{a_0}{3} \bar{\rho}_b(< r) = \frac{a_0 M_b(< r)}{4\pi r^3}, \quad (1)$$

a charge $a_0/4\pi G = 53.4 M_\odot \text{ pc}^{-2}$ per unit enclosed baryonic mass, half of the $106.9 M_\odot \text{ pc}^{-2}$ column ceiling. For a point mass with $x = r/r_M$: $M_c = M(\sqrt{1+x^2} - 1)$, $P = a_0 M/8\pi r^2$ and $\sigma^2 = V_c^2/2$. For extended baryons ρ_c depends only on $M_b(< r)$, and the target is equivalent to a locally virialised collisionless fluid with anisotropy $\beta = -\frac{3}{2}\rho_b/\bar{\rho}_b$. An additive law (phantom plus a cosmic-share cold fluid) overshoots by $+0.43 \text{ dex}$ at $x = 3$ and $+0.68 \text{ dex}$ at $x = 1$. With the fluid feeling the Newtonian potential of all mass, hydrostatics is an identity, so the whole content of the target is the closure. The point-mass identities are certified in Lean (24 theorems, Section 6).

What was excluded, with the exact hypothesis of each. Scripts were written by a delegated agent and re-run by the programme’s hub; all main runs pass and all controls fail as required (CFG44: 40 checks and four controls; CFG50: 12 checks and two controls). Hypotheses for the whole class: spherical, static, Newtonian equilibrium (the GR correction is at most 3×10^{-4} for $M_b \leq 10^{12}$), GR plus real mass.

mechanism	exact reason it fails
universal barotropic $P(\rho)$ in g_{tot}	the required c_s^2 scales as M^e at fixed density with $e = (2x^4 + 4x^2 + 1)/(4x^4 + 4x^2 + 1) \in [\frac{1}{2}, 1]$; the spread over $M_b = 10^8\text{--}10^{14}$ is at least 10^3 ; the effective index runs from 2 to 1
a second potential or fixed-kernel force linear in M_b	the deep regime forces $\beta = -1$ and $P \propto \rho^3$; the Newtonian regime then forces the felt force to zero
local closures (constant charge, $P = a_0 g_N / 8\pi G$, field-energy P , $\sigma^2 = r\nu g_N / 2$)	far-shell theorem: identical local fields, yet P differs by $(a_0/2) m / (4\pi R'^2)$ (+65% for a $10M_b$ shell at 40 kpc); off by 0.3–1.4 dex
adiabatic maintenance	adding baryons outside R' changes no force inside yet raises P
a Lagrange-constraint action	under GR plus real mass it is redundant with the barotropic case or overshoots (M_λ/M_c up to 3.4)
coupling to the phantom potential	reaction on the baryons 0.11–1.5 g_{law}
density slaving	reciprocity: the reaction on the baryons is inward, 1.2–4.2 g_{law}
a state-independent field-energy stress	$\omega^2 = -igk$: Hadamard ill-posed
tidal-tensor coupling (CFG50): stress = $\frac{1}{2}\chi T_{ij}\Pi^{ij}$ with T_{ij} the tidal tensor of a baryon-sourced auxiliary potential	T_{ij} is divergence-free, so as a stress it exerts no force; by reciprocity the reaction on the baryons is $O(g_{\text{law}})$ (at $r = h$: 1.3, 0.54, 0.03 for $M_b = 10^9, 10^{10}, 10^{12}$); the largest ghost-free outward force on the fluid is 0.505 g_{tot} , mass-independent; the needed χ differs by a factor 64 across $10^9\text{--}10^{12}$; in FRW it renormalises the fluid inertia by 50–3400 at $z = 1100$ unless Φ_b is built on the overdensity (postulated)

Two constructions survive, both as restatements: a temperature-slaved fluid (it postulates $\sigma_\infty^2 = \frac{1}{2}\sqrt{Ga_0 M_b}$, the baryonic Tully–Fisher relation, the cusp amplitude and the nonlocal exterior term) and a locally virialised collisionless fluid with $\beta = -\frac{3}{2}\rho_b/\bar{\rho}_b$ (the Jeans identity is derived, β is postulated, and positivity of $f(E, L)$ is open). Not tested: anisotropic $f(E, L)$ positivity; non-spherical baryons beyond an estimate (a thin disc’s tidal closure and enclosed-mass form differ by a factor 3–15); any relativistic completion. The tidal test assumes the coupling must carry the full closure force.

The sharpest statement. The single function of radius

$$C(r) = \rho_c r^3 g_{\text{tot}} = \frac{a_0}{4\pi} M_b(< r) \quad (2)$$

has no dynamical origin in any construction tried. It is not an equation of state, a linear or Poisson coupling, a local function of the fields at r , a state-independent stress, a constraint under GR plus real mass, a coupling to the phantom potential, a density slaving, a tidal-tensor coupling, or adiabatically maintainable. What is left is a non-adiabatic, nonlocal exchange of energy between the baryons and the fluid that depends on the enclosed baryonic mass. The baryonic Tully–Fisher relation is the postulate.

The reduction of Gap 2 to Gap 1. An enclosed-mass, non-adiabatic exchange that acts only where the system is bound and is set by the system’s own baryons is the same kind of object as a bound-only, owned switch: both are functionals of the baryon history of a top-level system, not of the local instantaneous state. The committed maps therefore count the remaining gaps as two, a legal bound-only owned switch and the tie, and state that Gap 2 “still reduces to Gap 1” after the tidal test. We flag this as an identification made in the maps from the form of the missing object; it is not a theorem, and a construction that produced (2) without a bound-only switch would falsify it. CFG48 reached the same identification from the Gap-1 side: Gap 1 and Gap 2 collapse into one object, the cold fluid’s dispersion (or stress) set by the enclosed baryonic mass at the edge. As a field-only gate it is impossible (Gauss); as an exchange action its reaction on the baryons is of order or larger than the law itself; as a history it is either acausal inside an action or a prescribed label. The surviving readings are all of the form “the fluid’s state is initial data from collapse”, which is what CFG44’s temperature-slaved and locally virialised survivors already were. This is still an identification, not a theorem.

Is the dispersion set by dynamics anywhere? (CFG60). A documentary lane asked, with the rubric frozen before its table was built, whether any committed construction gives the fluid’s stress $C(r)$ from the enclosed baryonic mass through a dynamical mechanism, as opposed to a postulate or collapse-time initial data. Each of 54 committed constructions received exactly one class: D (derived from an action or field equations with no postulate of the target), P (postulates the target or an equivalent), I (initial data), X (excluded by a committed script) or U (untested). Tally: D=0, P=11, I=5, X=27, U=11. Table B (6 rows) holds closure-map N-rows that do not bear on the dispersion; without it U=5. The answer is a scoped NO: no committed construction is class D. It is not

a proof of impossibility, it holds under the hypothesis boundary of each excluded class, and it is documentary: it re-runs no physics. Its audit script re-reads each cited check (122 of 122; re-run by the hub).

4 The tie: one action for $a_0(\Lambda)$ and a conserved fluid (CFG43)

The construction (units $c = 1$, $M_P^2 = 1/8\pi G$, $\epsilon = \kappa^2/8\pi$, $\kappa = \frac{1}{2}$ fitted) is

$$S = \int d^4x \left\{ \sqrt{-g} \left[\frac{1}{2} M_P^2 R - M_P^2 \Lambda \right] + M_P^2 \Lambda \partial_\mu T^\mu - \sqrt{-g} \rho(n; \Lambda) + J^\mu \partial_\mu \theta \right\}, \quad (3)$$

with Λ a Henneaux–Teitelboim [HT89] field with multiplier T^μ , a Schutz–Sorkin [SS77] irrotational fluid with one conserved current, and the cap entering as $\rho(n; \Lambda) = mn + P_{\text{cap}} x \arctan x$, $x = mn/(\nu_* M_P^2 \Lambda)$, $P_{\text{cap}}(\Lambda) = \epsilon M_P^2 \Lambda = a_0^2/8\pi G$. That gives $P = P_{\text{cap}} x^2/(1+x^2) \leq P_{\text{cap}}$; with $\kappa = \frac{1}{2}$, $a_0 = \sqrt{8\pi G P_{\text{cap}}} = 9.3603 \times 10^{-11} \text{ m s}^{-2}$, so the kernel and the fluid’s cap read the same constant.

Derived by the scripts, from the action (11, 5 and 11 checks in the three main runs; all six controls fail as required): Λ is a global integration constant; number conservation; Friedmann equals GR plus Λ plus dust; the Dirac count on a lattice (fluid alone $2N$; adding the HT sector $2N + 2$ with $2N - 1$ first-class constraints and no tertiary one, so the local degrees of freedom are unchanged and there is one global pair); $a_0(z)$ flat to 6×10^{-11} for $z \leq 10$; Ω_c is initial data; a column bound $\Sigma \leq a_0/2\pi G$ ($106.9/129.2 M_\odot \text{ pc}^{-2}$). **Postulated:** the tie function $P_{\text{cap}} = (\kappa^2/8\pi) \rho_\Lambda c^2$ (so it agrees with the earlier kernel-only tie by construction); the saturating shape and its argument; irrotational flow; a hard bound; minimal coupling. **Fitted:** $\kappa = \frac{1}{2}$; Ω_c ; and a new dimensionless constant ν_* . **Open:** a cap that binds across the baryonic Tully–Fisher mass range; shell crossing; the galaxy law itself; the bound-only switch; T5’s max rule; PPN.

The scoped obstruction. Hypotheses: a single conserved current; an equation of state barotropic in $\nu = \rho/\rho_\Lambda$ with $P \leq P_{\text{cap}}$; linear growth close to Λ CDM; the cap reached at r_M . (1) At $\nu_* = 1$, $c_s^2 \approx 6 \times 10^{-3}$ today and linear growth is 5–14% of Λ CDM. (2) Growth within 5% needs $\nu_* \geq 2 \times 10^4$ to 1×10^7 at $k = 0.5\text{--}30 \text{ Mpc}^{-1}$. (3) The density at r_M in units of ρ_Λ scales as $M_b^{-1/2}$ exactly, and the growth bound at the halo’s own scale scales the same way, so $g_0 = \nu_M/\nu_{\text{min}}$ is a pure number and one ν_* engages the cap over a mass window g_0^2 whatever ν_* is: direct re-solves at 10^9 and $3 \times 10^{11} M_\odot$ give $g_0 = 1.08$ and 1.04 ; over convention choices g_0 ranges over $0.08\text{--}3.3$, so the window is *at most about* $11\times$ in mass, against the 10^4 the baryonic Tully–Fisher relation needs. This is a screening estimate (two-fluid Newtonian growth).

The referee’s caveat, which limits how this is read. An independent re-implementation with its own growth solver reproduced the obstruction ($\nu_{\text{min}} = 2.2 \times 10^4$, 1.8×10^5 , 2.0×10^6 , 1.0×10^7 at $k = 0.5, 2, 10, 30 \text{ Mpc}^{-1}$; the lane’s values differ by 5–12%; the exponent 1.49–1.50; $g_0 = 0.36/0.355$ at the 5% criterion, $1.06/1.04$ at 20%). But “at most about $11\times$ ” is too strong as a ceiling: it holds only if suppression beyond 50% is disallowed and $F \geq \frac{1}{2}$ is kept; stacking the generous choices reaches $10^4\text{--}10^5$, each individually unmotivated, and $F \geq 0.01$ contradicts the hydrostatic requirement. Other cap shapes were not tested. The obstruction stands as stated for the saturating barotropic equation of state; it does not touch a fluid whose cap is not a barotropic function of its own density, for example an order-parameter fluid.

Status. Gap 3 is partly filled: the tie can act on a conserved fluid in one action without adding local degrees of freedom. The cap’s entry cannot be derived, and the natural saturating form needs a new constant. This is an existence result and a scoped no-go, not a mechanism. In addition the P2 point-mass pressure exceeds this cap for $r < r_M$ (its $P/P_{\text{cap}} = g_N/a_0$), so the fluid of this section and the fluid of eq. (1) are not one fluid.

5 The cold-mass rule on the dwarfs and the massive early types (CFG42, CFG45, CFG46, CFG51, CFG55, CFG58, CFG59, CFG64, CFG65)

The rule. The cold fluid is conserved, so a bound system keeps the cold mass it collapsed with: the dark mass is the law’s phantom plus leftover collapse debris. In lanes CFG42–CFG51 the rule adds $G f_{\text{ex}}(1 - f_b) M_{\text{NFW}}(< r)/r^2$ to the law’s acceleration, with the collapse mass from the Moster et al. relation [MNW13] and, where measured, from weak lensing [M16]. With weak-lensing collapse masses the rule leaves every star-forming spiral and passive disc below $\log M_* = 11.2$ on the law and fits the SLUGGS massive early types at 0.4σ against the law’s 3.3σ deficit

(CFG35–CFG38), with SLUGGS’s own population stellar masses; that pass did not survive dynamical stellar masses (CFG55, below). It is bookkeeping derived from T4, not a variation of an action: in Lean it is certified only as algebra of the rule as stated.

Ultra-faints and classical satellites (CFG42). Canonical footing; law and rule, offset in dex (significance):

population	the law	the rule
MW ultra-faints (Kaplan–Meier, limits in)	+0.325 (3.77 σ)	−0.059 (−0.41 σ)
MW classical dSphs	+0.027	−0.118 (−1.78 σ)
M31 Collins+13	+0.064	−0.024 (−0.22 σ)
M31 LVD	+0.044	−0.107 (−2.67 σ)

The headline (the ultra-faint offset is closed) passed; the classical-satellite check failed (H2). The collapse-mass floor on the ultra-faint result is 0.133 dex; the offset crosses zero near $3 \times 10^8 M_\odot$, three times below the clamped 10^9 . *The referee’s caveat:* the closure is a weak test. Every collapse mass from 2×10^8 to $10^{12} M_\odot$ passes the 2σ criterion, only masses below about 3×10^7 fail, and the significance falls from 3.8 to −0.4 partly because the error grows from 0.086 to 0.143; the clamp covers 33 of the 40 ultra-faints. The rule trades B’s largest failure for a smaller one in σ (1.8–2.7 σ), spread over more objects; this is the cusp problem reached from B’s side. The debris would have to switch off where the phantom already supplies the observed mass: a design constraint on T5’s max rule (dark = the larger of the phantom and the cold budget, not their sum), *not* a fix.

Four readings, one conjunction (CFG45). Four pre-declared readings of “the fluid keeps its collapse mass” were scored on every population with the lanes’ exact statistics (66 committed numbers reproduced): (L) the bare law, (S) the sum as committed, (M) the radial max and (E) the exterior leftover. Acceptance is the conjunction of seven gates. **No reading passes all seven.** (S) passes 6/7 and fails only the classical satellites (M31 LVD −2.67 σ) among CFG45’s seven gates, with SLUGGS’s own stellar masses (later lanes add SLUGGS with dynamical masses at 2.6 σ , CFG55 and CFG71, and the Local Volume field dwarfs at −3.47 σ , CFG58); (M) passes 4/7 and fails the classical satellites, SPARC (where 36% of the massive spirals move by more than 0.03 dex) and SLUGGS (−2.12 σ); (L) passes 5/7 and fails the ultra-faints (3.8 σ) and SLUGGS (3.3 σ); (E) passes 5/7, fails the same two, and is identical to the bare law in every lane, because the exterior shell never reaches a data radius. Control C2a failed as declared and is kept.

A statistical binary correction (CFG46). Arroyo-Polonio, Battaglia & Thomas [AP26] correct single-epoch dispersions with a mixture model. Eight of their twelve ultra-faints are scorable. The bare law’s median offset is $+0.295 \pm 0.163$ (1.81 σ) at $f = 0$, $+0.206 \pm 0.164$ (1.26 σ) with f free, and $+0.171 \pm 0.172$ (0.99 σ) at $f = 0.7$. The correction lowers each dispersion by −0.068 to −0.104 dex, a quarter to a third of the offset, and seven of the eight offsets stay positive; but with eight systems against 40 the test lost power, so the pre-declared H1 and H2 failed largely for that reason. It neither confirms nor removes the failure. The rule with the corrected dispersions sits at −0.151 (−1.17 σ).

Multi-epoch, binary-cleaned dispersions (CFG51). From per-epoch velocities (Walker et al. [W23]; 49,367 rows, 38 systems) only two ultra-faints are informative. Cleaned offsets: Boötes I $+0.219 \pm 0.089$ (2.46 σ) and Tucana II $+0.465 \pm 0.128$ (3.63 σ); cleaning lowers the offset (single-epoch +0.310, +0.730) but does not remove it; the sum rule over-predicts both (−0.205, −0.165). The data lie between the bare law and the sum rule. Caveats: Boötes I’s cold component alone (2.4 km s^{−1}, from the literature) would give +0.007, so its result rests on the hot component (CFG66, next, measures the cold component alone at -0.07 ± 0.20 instead, consistent with no failure); Tucana II is tidally disturbed (CFG66 tests a velocity gradient); membership was re-derived, and control C1 failed as declared (Tucana II single-epoch 7.48 against a literature value of 3.8, which is the binary inflation itself). Two objects are not a population result.

The two flagged systematics (CFG66). A follow-up asks whether the Boötes I and Tucana II offsets survive (a) a two-Gaussian mixture for Boötes I and (b) a linear velocity gradient for Tucana II, where survival means the offset stays positive above 2σ on both footings. (a) Boötes I (53 cleaned stars): the fit gives $s_{\text{cold}} = 2.00$ and $s_{\text{hot}} = 5.09$ km s^{−1}, but the split is not detected (likelihood-ratio improvement 2.31, bootstrap $P = 0.155$). The total-mixture offset is $+0.222 \pm 0.097$ dex (2.29 σ ; alt +0.202, 2.09 σ), a marginal pass; the cold-only offset is -0.072 ± 0.203 (−0.35 σ), consistent with zero, and replaces CFG51’s “+0.007”. (b) Tucana II (12 cleaned stars): the gradient is 0.43 km s^{−1} per half-light radius, with an error of about 3.4 (likelihood-ratio $p = 0.99$), and the offset is unchanged at $+0.464 \pm 0.128$ (3.62 σ). With 12 stars a moderate gradient cannot be excluded; the Boötes I margins are thin (the alt footing barely clears 2σ) and the data cannot say whether its hot component belongs to the galaxy. Two objects are still not a population result. A first-run normalisation bug was fixed before any result

was used; the MUTATE run fails S1 and S2 as required; the lane was written by a delegated agent and re-run by the hub.

SLUGGS with dynamical stellar masses (CFG55). The rule’s one clean pass, SLUGGS, used SLUGGS’s own population stellar masses, and the standing objection was the stellar mass. CFG55 instead fixes each model’s stellar mass from the galaxy’s own inner kinematics (the ATLAS3D JAM mass inside the half-light sphere; 16 of CFG38’s 19 galaxies, the other three lying outside ATLAS3D’s declination range) and then predicts the outer globular-cluster dispersions, so no IMF enters. Canonical: the law under-predicts by $+0.097 \pm 0.024$ dex (4.0σ ; alt 3.65σ) and the rule by $+0.046 \pm 0.018$ (2.6σ); with SLUGGS’s masses on the same 16 galaxies the rule had left $+0.007$ (0.4σ). Both declared hypotheses failed: the law fits (H1) and the rule fits (H2). An independent re-derivation (CFG76) reproduces these offsets exactly, and shows post hoc that they are statistical errors at a fixed globular-cluster density slope $\gamma = 3$ for every galaxy: the law’s offset is zero at $\gamma \approx 1.8$ and the rule’s at $\gamma \approx 2.5$, and without the four group and cluster centrals the law is $+0.055$ (2.7σ) and the rule $+0.026$ (1.3σ). The calibration lowers the law’s stellar masses (median -0.10 dex), because the law’s own phantom supplies part of the mass inside the half-light radius, so the deficit grows and the IMF objection runs the wrong way. Control C3 failed as declared (a rounding constant) and is kept. Unmodelled: hot gas (three of the four worst offenders are X-ray-bright group or cluster centrals) and globular-cluster orbital anisotropy. The lane’s own reading: the rule no longer fits cleanly anywhere among massive passive systems.

More populations (CFG58). The sum S switches itself off when $f_{\text{ex}} = 0$, that is where the law’s edge phantom already exceeds $(1 - f_b)M_c$. That holds above $M_b \approx 2.3 \times 10^7 M_\odot$ (canonical; 1.5×10^7 alt) for $M_* = M_b/2$, so every population heavier than that is unchanged by the rule, by its own arithmetic. What the rule breaks are the lowest-mass Local Volume dwarfs: for the 13 LV field dwarfs the median goes from -0.60σ (law) to -3.47σ (sum, in the sum’s own error 0.030) canonical, -0.85σ to -2.70σ alt, a change of 0.061 dex that is only -1.4σ in the law’s own error; the same signature as CFG42’s classical-satellite failure. The blind, non-exempt versions for NGC1052-DF2/DF4 and the tidal dwarfs fail under both the law and the sum, so the rule adds nothing there. Control C1 failed and is kept: one comparison differs by $2\text{--}5 \times 10^{-9}$ against a tolerance of 1×10^{-9} because the campaign footing uses the rounded a_0 pair, and a reported diagnostic with the exact a_0 reproduces it.

What this means for an action. No reading of the rule closes B, and the maps’ only constraint is qualitative: a partial debris that vanishes where the phantom supplies the mass and survives where it does not. A fitted form of that constraint is a knob; the programme’s rule is to derive it. No action yields it. The lanes since then add three checks of how much the rule’s dwarf results depend on our own choices; they are collected next.

5.1 Robustness of the rule’s dwarf results

Kernel (CFG64). Is any headline verdict of CFG40, 41, 42, 45, 46, 51, 56, 58 and 59 sensitive to the interpolating kernel, ν_{mono} (the campaign default) against P2 ($\nu = \sqrt{1 + 1/y}$; CFG14 found P2 disfavoured at about 2σ by the SPARC transition shape, not excluded)? The question was frozen first and the harness swaps the kernel at code level in every lane. Answer: every headline is kernel-robust. The largest move anywhere is 0.027 dex and 0.39σ , and the ultra-faint lanes move by at most 0.012 dex (CFG42: rule $-0.059 \rightarrow -0.058$, law $3.77\sigma \rightarrow 3.90\sigma$; Boötes I $2.46\sigma \rightarrow 2.52\sigma$). Of the 22 declared hypotheses exactly one is kernel-sensitive, CFG56 H1, which crosses the 2σ line by a 0.027 -dex move; it is not a headline. Two disclosures about our own text. First, CFG42, CFG46, CFG51 and the FG001-derived parts of CFG45, CFG58 and CFG59 used the exponential RAR kernel $\nu = 1/(1 - e^{-\sqrt{y}})$, not ν_{mono} as their docstrings and READMEs said; the two agree to 3.3×10^{-9} for y from 10^{-4} to 0.1 (the ultra-faint regime) and to 2.3% up to $y = 30$, so the numbers stand and the description error in six lanes’ docstrings is corrected in the READMEs. Second, “P2” here means the kernel swapped with everything else frozen, not a self-consistent refit: the SPARC stellar Υ came from a ν_{mono} fit and was not refit. The control C3 declared before the first run was badly chosen and failed as declared, and is kept; C3b, added after seeing C3 and disclosed, passes.

Halo shape (CFG65). The debris uses an NFW shape with the Dutton–Macciò concentration, which reviews flagged as the fragile assumption behind both the ultra-faint closure and the classical-satellite over-prediction. Three parameter-free alternatives were declared before the first run: (A) NFW with Duffy et al. concentrations (A1: the committed 200m row of CFG23; A2 and A3: the 200c rows), (B) a Burkert core at $r_{\text{core}} = r_s$, (C) Einasto with $\alpha = 0.17$. Canonical offsets in dex, ultra-faint Kaplan–Meier median then M31 LVD: committed NFW -0.059 and -0.107 ; A1 -0.098 and -0.125 ; A2 $+0.049$ and -0.071 ; A3 -0.007 and -0.096 ; Einasto -0.117 and -0.124 ; Burkert $+0.284$ (3.91σ) and -0.006 (-0.09σ). NFW under all three concentration relations, and Einasto, leave the ultra-faint offset under 0.75σ and move the LVD by at most 0.036 dex; under A2 the LVD significance drops to -1.7σ (alt -1.9σ), below the -2σ line, and passes only through the 0.05 -dex clause. A Burkert core is shape-

sensitive on both results: it leaves the ultra-faints at $+0.284\text{dex}$ and removes the LVD over-prediction, while SLUGGS becomes a $2.0\text{--}2.6\sigma$ under-prediction. *The debris result is robust to the concentration relation and to an Einasto profile, and it is not robust to a core.* A cusp closes the ultra-faints and over-predicts the classical dwarfs; a core removes both. The closure needs the debris mass at $0.03\text{--}0.3\text{kpc}$ to be at least about 0.55 of the NFW value. Burkert with $r_{\text{core}} = r_s$ and Einasto with $\alpha = 0.17$ are single declared choices, not a claim that all cored profiles fail. Caveats: the Duffy coefficients of all three rows were first transcribed from memory and were later compared by a second reader with Table 1 of Duffy et al. (arXiv:0804.2486, from the paper text) and match exactly; the published MNRAS version was not compared. A1 is a 200m row applied to a 200c mass and is about 1.75 times A2 at every mass, so the convention, not the transcription, is the dominant systematic: A2 and A3 are the consistent rows, and they give the LVD over-prediction at -1.7σ (alt -1.9σ , A2) and -2.3σ (A3), robust in size and marginal in significance. The multi-epoch lane (CFG51) was not run.

No universal debris fraction (CFG59). A frozen consistency question: scale the sum’s debris acceleration by $\varphi \in [0, 1]$ ($\varphi = 0$ the bare law, $\varphi = 1$ the sum) and ask whether all ten populations’ 1σ intervals of φ share a common intersection. The answer is NO on both footings at the frozen 1σ acceptance; at 2σ a common φ exists for these ten populations (canonical $[0.52, 0.71]$, alt $[0.33, 0.66]$), so the verdict is a statement about tight consistency, not about any fraction being excluded at 2σ . The binding conflict is SLUGGS ($\varphi \geq 0.81$) against the M31 LVD ($\varphi \leq 0.30$), a gap of 0.51 (alt 0.46); the ultra-faint median needs $\varphi \approx 0.74$ (interval from 0.39) and the classical dSphs $\varphi \leq 0.39$. Dropping the ultra-faints, SLUGGS and the X-ray ellipticals leaves $[0.22, 0.30]$ (alt $[0.21, 0.22]$): the low-mass satellites want about a quarter of the debris and the most massive early types want nearly all of it. The main run passes 5 of 5 and MUTATE fails the headline as required. CFG65 adds that the LVD and SLUGGS intervals stay disjoint for DM, A1, A2, A3 and C, that is for the committed NFW, the three Duffy rows and Einasto. With SLUGGS’s dynamical stellar masses (CFG71) the picture changes: no $\varphi \in [0, 1]$ brings SLUGGS within 1σ ($\varphi = 1$ halves the offset, $+0.097$ to $+0.046\text{dex}$, and 2.6σ remains), so SLUGGS needs more than all of the sum’s debris; the binding pair among the populations that still have intervals is the ultra-faints against the Local Volume (gap $0.087 \mid 0.164$), and the lane’s pre-declared classification printed MIXED (disclosed there). The universal- φ answer stays NO at 1σ ; at 2σ the intersection is empty only through SLUGGS (2.6σ at $\varphi = 1$; dropping SLUGGS alone gives $[0.23, 0.71]$). This is a statement about the sum’s mass dependence, not a rule, and no φ is selected. Caveats: the clamped Moster relation and the NFW cusp, the lane error models, Boötes I’s hot component, and the possibly tidally inflated Tucana II.

6 What Lean certifies, and what it cannot

The composable Lean core is `fable_independent_2026/lean_2026/ChainCert`. Its verifier checks that it builds, has no `sorry`, and that all 161 theorems depend only on `propext`, `Classical.choice` and `Quot.sound`; the verifier’s mutation control adds an unproved theorem and fails as required. **What Lean certifies here is that the conclusions follow from the stated premises; it certifies no empirical fact.** $\kappa = \frac{1}{2}$ is fitted, Ω_c is fitted, the constancy of ρ_Λ is a premise, and the kernel is declared. Since the first version of this note the library has gained a `Profile` module (the extended-profile identities of CFG44), a `Cusp` module (the halo-cusp scaling of CFG65), a `Gauss` module (the algebra of CFG48’s solution check), a `Separation` module (the sample-size algebra of CFG63), an `Exchange` module (the algebra of CFG72’s mediator, including the exact trade-off $\delta f_{bb} = 1$) and its earlier modules were tightened by a semantic audit of the library.

link	what is certified, and its scope
form of a_0	the monomial $c^\alpha G^\beta \rho^\gamma$ has the units of an acceleration iff $(\alpha, \beta, \gamma) = (1, \frac{1}{2}, \frac{1}{2})$, both directions; certified, conditional on the monomial family
BTFR zero point, kernels	any kernel with $\nu(y)\sqrt{y} \rightarrow 1$ gives $v^4 \rightarrow GMa_0$; holds for ν_β and ν_{mono}
tie and flat $a_0(z)$	flatness is the premise $\hat{h}_{\text{flat}}$ (constant ρ_Λ , empirical) propagated; without it the conclusion is false
rival $a_0 \propto H(z)$	$E(z) > 1$ and increasing; $E(2.5) \in (3.76, 3.761)$ at $\Omega_m = 0.3138$
cold-mass rule	$f_{\text{ex}} = \max(0, 1 - M_{\text{ph}}/M_c)$ gives $M_{\text{ph}} + f_{\text{ex}}M_c = \max(M_{\text{ph}}, M_c)$: algebra of the rule as stated, not derived from an action
fluid cap (CFG43)	$P = P_{\text{cap}}x^2/(1+x^2) \in [0, P_{\text{cap}})$ and $a_0^2 = \kappa^2\Lambda/8\pi$: an identity inside the fluid module, <i>not linked</i> to the kernel's a_0 in Lean; the cap's entry form is a postulate
fluid = phantom, point mass (CFG44)	exact real-analysis identities for the explicit- $\sqrt{}$ P2 law (24 theorems); Lean does not connect it to ν_β at $\beta = 1$; no_single_polytrope is stated for $\Gamma(x)$ and needs one extra step for the effective-index form; the target itself is a declared law
fluid = phantom, arbitrary spherical profile (CFG44; module Profile)	for any $M_b(r)$: $d/dr [a_0 g_N / 8\pi G + (a_0/2)\Sigma_{\text{out}}] = -a_0 M_b / (4\pi r^3)$; given the target, hydrostatic equilibrium holds for every profile; with $\Sigma_{\text{out}} \rightarrow 0$ the pressure is unique and positive; $\sigma^2/(V_c^2/2) = 1 + 4\pi r^2 \Sigma_{\text{out}}/M_b$; the target is equivalent to the ODE $GM'_c = a_0 r u_N/u$. Certified as identities; the target is a declared law. No extended profile satisfying both the decay hypothesis and the target was constructed, only the point mass: an isothermal profile witnesses the target without decay, an exponential profile witnesses the decay without a cold fluid; ODE solvability, $\rho_c \geq 0$ on real profiles and Eddington positivity are numerical
κ	the zero point is linear in κ by cancellation; the stated relation is onto, and that Z_q is free is a premise, so $\kappa = \frac{1}{2}$ is tuned, not derived

Not certified. Ownership and the bound-only switch (no formal statement exists without a written specification); the action itself (nothing formalises one); the relativistic sector (slip, PPN, Shapiro, stability beyond scoped certificates); the uniqueness of the form of a_0 over all dimensionally allowed forms (only the monomial family is covered); every empirical claim. The wider corpus is 137 files, of which 116 compile, 21 fail, and 9 real **sorry** occur in 4 non-certificate files; the compiling files are islands (no file imports another) apart from ChainCert, the first library whose theorems import each other and compose into a conclusion. The corpus's per-file descriptions were not independently re-verified.

Corrections from the semantic audit. The library's README states its scope limits, and the audit commit applied them: flatness of $a_0(z)$ is the premise propagated, with no content beyond “ ρ_Λ constant implies a_0 constant”; C1's “iff” is now proved in both directions (**C1_a0_form_iff**); the κ row states that the relation is onto and that Z_q is free is a premise; **zero_point_linear_in_kappa** is a cancellation; **cap_a0_tie** is not linked to the kernel's a_0 ; the point-mass P2 pressure exceeds the cap inside r_M , so the Fluid and PointMass modules are not one fluid; and **no_single_polytrope** is stated for $\Gamma(x)$. The audit found no unsound theorem, no vacuous hypotheses, standard axioms only. The closure map's equation ledger records an independent re-run of its rows 1–11 (26 runs, main and every MUTATE control) that reported 0 mismatches in exit codes and check counts; the report itself was uncommitted at the time the ledger was written.

7 What decides the rest

Gaia DR4 wide binaries, 2 December 2026. The pre-registration is frozen, with Amendments 13, 14 and 15 filed. The test cuts both ways among the framework's own readings; every “dies” applies only if the frozen stability checks pass.

reading	prediction for $\hat{\gamma}$ (can/alt)	dies if
Arm A: the kernel as strict MOND, with the external field	1.16–1.18 / 1.19–1.23	$\hat{\gamma}$ Newtonian
Arm C: hierarchical ownership (B's rule)	1.000 exactly	$\hat{\gamma} \geq 1.084$
the derivation chain's law (Amendment 14)	ceiling 1.0725 / 1.0900	$\hat{\gamma} \geq 1.157/1.174$

A MOND-like boost kills the ownership rule and the chain's law; a Newtonian result kills strict MOND, and leaves B standing but unconfirmed over Newton.

a_0 at $z \approx 2.5$. The flat law predicts a zero shift in the galaxy zero point and Λ CDM about +0.334 dex; at 0.13 dex per object the flat law is 2.57σ from the Λ CDM-native value, and 1.5 – 2.4σ if an evolving dark-energy branch holds

(branches move $a_0(2.5)$ by -0.10 to $+0.14$ dex). The registered test needs 3 rotators at ± 0.10 dex or 4 at ± 0.20 . *It cannot be run on the data on disk* (CFG52, a feasibility count and not a test of B): at $z \geq 1.5$ no clean object has $g_{\text{bar}} < 0.3 a_0$; of 242 galaxies with $g_{\text{bar}} < a_0$, 16 have a prediction gap between the flat law and $a_0 E(z)$ above 1σ , none above 2σ ; pooled at $z \geq 1.5$ ($N=14$) the flat law sits at $+0.135$ and the rival at -0.037 dex ($+1.0\sigma$ and -0.3σ with a correlated mass calibration) and a selection mock absorbs most of it. MUSE-DARK II at $z \approx 1.2$ favours the flat law by about 1σ but is gas-dominated, and MUSE-DARK III's rising a_0 to $z \approx 1.4$ is not diagnostic (Λ CDM simulations produce the same apparent rise). A decisive test needs 2–4 JWST integral-field plus ALMA discs at $z \approx 2.5$ with the mass calibration below about 0.1 dex; a correlated 0.2 -dex systematic caps the significance near 2.3σ against $a_0 \propto H(z)$ and about 1.3σ against Λ CDM-native (1.7σ on a different floor reading; CFG52, CFG63, where the bridge is derived and not committed) whatever the number of discs; the 2.57σ above is one object, statistical error only.

The PHIBSS check (CFG54). A pre-registered scoring of flat a_0 against $a_0 E(z)$ on the Tacconi et al. PHIBSS sample (73 galaxies at $z \approx 1.2$ and 2.2 ; 51 remain after excluding the 3σ gas upper limits and the rows whose quoted gas fraction their masses do not reproduce), restricted to discs with $g_{\text{bar}} < a_0$ at the velocity radius and gated by feasibility ($N \geq 5$). Result: $N = 0$ of 51. The lowest g_{bar}/a_0 at $r_v = 1.31 r_h$ is 1.10 (median 4.7), so no disc is in the regime where a_0 shows, and the lane is empty and non-diagnostic; that is the declared, valid result (the main run exits 1 on the feasibility gate, and its MUTATE control is vacuous, as disclosed). The answer hinges on a radius the table does not carry. Post hoc, reported and not a test: the numbers of the 51 that would qualify are 0 at $1.31 r_h$, 7 at $2 r_h$, 20 at $3 r_h$ and 31 at $4 r_h$; choosing the radius after seeing the counts would not be pre-registered. This agrees with CFG52.

Which open test can discriminate (CFG63). A forecast from committed budgets only (65 citations checked by the script; no new physics or constant): only Gaia DR4 separates B from anything, and only from the bare MOND-type law (Arm A) (the ultra-faint row, capped at 2.5σ , separates the bare law from the derived rule, not B from a rival). Three sigma needs about 4,342 pairs (alt 2,940) against 30,000 expected. It cannot separate B from Λ CDM or Newton at any N , because all three predict $\hat{\gamma} = 1.000$, so a Newtonian outcome is the survival of a test B could have failed, not a confirmation. Every other test with a usable σ is capped below 3σ by a systematic floor whatever N : ultra-faint dwarfs 2.5 , a_0 at $z \approx 2.5$ 2.3 against $H(z)$ tracking and 1.3 against Λ CDM-native (the floor bridge is derived in CFG63, not committed), and X-ray groups at R_{500} 1.8 . The KiDS budget is not forecastable from committed budgets. The main run passes 11 of 11 checks and its MUTATE run exits 1. Separately, the criteria for scoring B's derived rule against the KiDS-1000 early/late lensing split were frozen and committed before any CFG61 scoring script exists (the file's own wording). The scored result is now committed (CFG61): the frozen amplitude statistic is degenerate (the rule adds nothing at KiDS lens masses) and its two hypotheses failed as coded; a post-first-run χ^2 row for the early-minus-late difference gives $28.1/7$ ($p = 2 \times 10^{-4}$, about 3.7σ), reported only, and a Λ CDM control with Mandelbaum halo masses fits the same difference (CFG67, χ^2 $6.5/7$; its own second hypothesis failed). It is not a scored rejection either way. An independent re-derivation (CFG77) reproduces the $28.1/7$ and $6.5/7$ exactly; it shows that the law's χ^2 is a pure data statement shared by any colour-blind model (the law's own predicted difference is negligible) and is fragile to modest unmodelled scatter (all errors $\times 1.5$ give $p = 0.086$), and that the hot-baryon ladder does not reproduce to its printed digits.

Multi-epoch ultra-faint data. The multi-epoch, per-star dispersions are the decisive test of the largest failure. Only two systems allow it today (Section 5), and CFG66 found that both total-mixture offsets survive their flagged systematics, Boötes I marginally (2.29σ , alt 2.09σ), while its cold component alone (-0.07 ± 0.20) does not, and the hot-cold split itself is not detected ($P = 0.155$); Hydrus I, Reticulum II and Ursa Major II have no repeat epochs in the catalogue used.

None of the three data sets would supply what an action must: they test B, not the existence of an action.

8 Limitations and what is not claimed

Limitations of this document.

- It is a compilation of committed material and introduces no new calculation. The gate list and the action catalogue were compiled by delegated readers and spot-checked by hand; their own headers say that most rows were not independently re-verified, and that a committed script output overrides them. The group and class counts of Section 1 are agent-derived recounts, re-run by the audit script.
- The CFG43, CFG44 and CFG50 scripts were written by a delegated agent and re-run by the hub; the extended-profile, barotropic, reciprocity and growth results are numerical, and were run, not re-derived. Every astra-lane

result is marked unreviewed and is not relied on. The later lanes CFG49, CFG58, CFG59, CFG64 and CFG65 were likewise written by a delegated agent and re-run by the hub; CFG48’s main runs were re-run by the orchestrating session and reproduced by a referee; CFG60 is a documentary count whose class assignments are the delegated agent’s, with an audit script that re-reads its cited checks.

- Input and scope caveats of the new lanes. The Duffy coefficients in CFG65 were checked against the preprint’s Table 1 but not the published version, the A1 row’s 200m-on-200c convention is the dominant systematic, and a Burkert core removes both the closure and the over-prediction, so the debris results depend on a cusp. CFG64’s “P2” is a kernel swap, not a refit. CFG55 omits hot gas and assumes isotropic globular clusters. CFG54’s outcome hinges on a velocity radius the source table does not carry. CFG48’s stability results are second-variation statements about a frozen background, and its ball gate’s first variation is not fed back. CFG49 has no frozen-criteria file (see above).
- The obstructions are scoped. Each excludes a class under its stated hypotheses and none is a theorem against an arbitrary nonlocal, multi-metric or explicitly anisotropic action. The reduction of Gap 2 to Gap 1 is an identification, not a theorem.
- The 42 of 48 is a recount, not an established score; the harness’s control K2 failed. The frozen list itself is unchanged; the later status file restates rows and is not a re-scoring of the harness.
- Both footings are quoted where the source gives both; where a source gives one, it is canonical. $\kappa = \frac{1}{2}$ is fitted; SPARC [LMS16] is the data behind the galaxy law; the a_0 – Λ link assumes Λ is a real cosmological constant (a de Sitter horizon).
- The referee reproduced the CFG43 obstruction with its own solver, but concluded that “at most about $11\times$ ” is not a ceiling.

Never claim (copied from the programme’s standing page):

- That dark matter is gone. The mass is required.
- That $\kappa = \frac{1}{2}$ is derived.
- That any data set favours the framework over Λ CDM.
- That the theory is closed or complete.
- Any Standard Model or theory-of-everything result. That sector is walled, and those claims were publicly retracted.
- That the Local Group or the budget tension excludes the framework (CFG23, CFG27).

9 Reproducibility

Every number quoted here is re-read from a committed file by `PAPER37_audit.py`, which parses this source for the marked values, checks each against its committed source at the precision quoted, reports how many match, and has a mutation mode that alters one expected value and must fail. The sources are `campaign_fresh_gravity/closure_map/`, `campaign_fresh_gravity/STANDING_2026-09-28.md`, `campaign_fresh_gravity/LEDGER.md`, `campaign_fresh_gravity/closure_map/GATES_STATUS_2026-09-29.md`, the READMEs and outputs of lanes CFG42, CFG43, CFG44, CFG45, CFG46, CFG48 (and its referee note), CFG49 (and its referee note), CFG50, CFG51, CFG52, CFG54, CFG55, CFG56, CFG58, CFG59, CFG60, CFG61, CFG63, CFG64, CFG65 and CFG66, and `fable_independent_2026/lean_2026/ChainCert`. Where a quoted statement is recorded only in a commit message (the CFG48 scope, the ChainCert audit outcome), the audit reads that message at git HEAD and says so.

References

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